

Detecting Fake Reviews Through Feature Engineering, Statistical Analysis, and Machine Learning

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Abstract—Real online reviews highly influence purchases, whereas untrue reviews sidetrack buyers and advertise inferior products. Fake review detection aims at the creation of features that are language-specific and/or semantic or behavior based, such as punctuation use, sentiment polarity, category relevance, etc. In the present paper which is focused on the fake review detection, feature selection conducted by Welch's t-test and chi-square is enhanced with correlation-based filtering to boost robustness. Clustering algorithms identified hidden patterns that are then added into the feature set for classification through Logistic Regression, Random Forest, XGBoost, and Gradient Boosting. Cross-validation measures are taken to evaluate the models by different categories. It is observed that combining classical statistical methods with machine learning algorithms can provide a scalable and interpretable solution to improving trust in e-commerce platforms.

Index Terms—Fake Review Detection, Text Classification, Sentiment Analysis, TF-IDF, Clustering, Logistic Regression

I. INTRODUCTION

The rapid rise of online marketplaces has placed user-generated reviews at the forefront of consumer decision-making processes. Reviews have emerged as a crucial alternative to local product inspections and, therefore, make an impacting difference in the purchaser's choice and the reputation of the seller. However, with the concern of fake reviews, which are posted deliberately by individuals, incentivized crowds, or automated bots, the matter concerning the trust, integrity, and fairness of e-commerce has assumed grim heights. Fake reviews serve to distort the perception of products, manipulate their ratings, and consequently exercise undue influence on consumer confidence and fair competition between legitimate businesses.

Current schemes attempt to detect fake reviews mainly through superficial features such as textual data, metadata, or behavioral signals. Although, able to detect fake reviews in a controlled environment, such methods now find it difficult to keep pace with the growing sophisticated nature of deceptive

texts and their natural imbalance between positive and negative classes.

In this paper, the proposed methodology systematically captures subtle deceptive cues from a wide spectrum of features that range from lexical, syntactic, sentiment, behavioral, and category-specific signals. The detection process is further enriched with unsupervised clustering to uncover latent structures in review data while the statistical validation is employed to select the most robust features. Per-category model training is then used in this system to offer higher adaptability and better generalization in various e-commerce domains.

II. LITERATURE SURVEY

Recent advancements in quantum computing have opened new possibilities for detecting fraudulent reviews in e-commerce. A quantum neural network-based approach has shown promise in handling large-scale datasets and outperforming conventional machine learning models in identifying deceptive reviews [1]. Earlier foundational work in this domain treated review spam as duplicated or near-duplicated content, combined with analysis of rating outliers, providing initial rule-based methods for spotting deception [2]. This was later enriched by incorporating behavioral footprints such as posting frequency, temporal patterns, and review consistency, thereby adding a critical social and temporal dimension to detection efforts [3]. The introduction of deep learning, particularly recursive neural networks, enabled models to capture complex semantic and syntactic structures in review text, surpassing traditional feature engineering methods [4].

Linguistic analysis also played a significant role, where features like function word usage, pronoun frequency, and markers of cognitive processes were used to differentiate fake from real reviews [5]. Hierarchical attention networks further refined textual modeling by attending to both global context and granular linguistic cues, allowing models to uncover subtle deceptive patterns [6]. Beyond textual analysis, matrix

factorization techniques were applied to exploit latent structures in reviewer-product interactions, exposing coordinated deceptive behaviors [7]. Semantic inconsistencies between the sentiment expressed in the text and the given ratings were also explored as strong indicators of manipulation [8]. Additionally, hybrid feature sets encompassing lexical diversity, part-of-speech tags, and n-grams proved more robust than single-feature models [9]. The importance of rigorous evaluation methods was emphasized through the development of balanced and well-annotated datasets for benchmarking [10]. Finally, research on domain adaptation techniques highlighted the need for models capable of maintaining accuracy across different product categories and platforms, thereby improving generalizability in real-world scenarios [11]. Combination of advanced natural language processing techniques along with ensembled models were implemented by [12] for the review anomalies. Examines video tampering technology with detailed discussions on the various detection methods. [13]. Aswin et al. [14] explores the use of deep learning techniques to detect and classify fake political and property-related reviews on social media, addressing the rise of toxic and misleading content.

III. METHODOLOGY

The fake review detection approach developed in the present study adopts a systematic, multistage methodology. First comes feature engineering, where the key linguistic and behavioral patterns are extracted from review text so as to embody its informative characteristics. Then the process of feature selection identifies and retains features with the highest discriminative power to separate between genuine and fraudulent reviews. From here, clustering techniques are performed so that some inherent structures and groupings could be discovered in the data, shedding more light on the distributions over reviews. Finally, given the refined feature set and the clustering solution, the classification models are then trained to predict the actual class of reviews. Each stage in the pipeline is crucial to building a resilient fake review detection solution.

A. Feature Engineering

Features are attributes(characteristics) or variables that describe the data. Features are used as inputs to make predictions or classify in machine learning models. Feature engineering is used to identify, select and extract the attributes that are most relevant from the raw data. The following are the features extracted:

- **category_keywords{}**: A dictionary where each product category (e.g., pet supplies, electronics, books) is a key, and its value is a list of keywords from reviews in that category.
- **category_relevance(text, category)**: Checks if the review text contains any keyword from the relevant category in the `category_keywords` dictionary. Returns 1 if a keyword is found, otherwise 0. This helps detect irrelevant or vague reviews.

- **get_entropy(text)**: Measures the randomness in character distribution using the entropy formula:

$$H(X) = - \sum_{i=1}^n p(x_i) \log_2 p(x_i) \quad (1)$$

where $p(x_i)$ is the probability of character x_i in the text. A higher entropy indicates greater randomness or lack of structure in the review.

- **count_punctuations(text)**: Counts all punctuation marks in the text. Overuse of punctuation (e.g., “!!!”, “...”) is often observed in exaggerated or fake reviews.
- **avg_word_length(text)**: Calculates the average word length in the text. Genuine reviews tend to use natural vocabulary, while fake ones often use overly simple or generic words.
- **sentiment_score(text)**: Uses `TextBlob` to compute a sentiment polarity score ranging from -1 (negative) to +1 (positive). Extremely polarized reviews may indicate artificial boosting or defaming.
- **uppercase_ratio(text)**: Measures the proportion of uppercase letters in the text. Excessive uppercase usage (e.g., “AMAZING PRODUCT”) can be a sign of forced emphasis.
- **exclamation_count(text)**: Counts the number of exclamation marks. Fake reviews often overuse exclamation points to create a sense of excitement.
- **question_count(text)**: Counts question marks in the review. Presence of rhetorical or promotional questions (e.g., “Looking for the best phone?”) may indicate templated content.
- **digit_count(text)**: Counts numeric digits in the text. Frequent or irrelevant digits (e.g., “Buy 3 now!”) are common in promotional or spammy reviews.
- **word_length_variance(text)**: Computes the variance in word lengths. Genuine reviews tend to have natural variation; fake ones often do not.
- **lexical_diversity(text)**: Calculates the ratio of unique words to total words. Low diversity suggests repetition or copy-paste behavior.
- **short_word_ratio(text)**: Measures the proportion of words with length ≤ 3 . Overuse of short, simple words (e.g., “so”, “is”, “good”) may imply lazy or low-information content.
- **repetition_ratio(text)**: Computes the ratio of repeated words to total words. High repetition may indicate use of templates or keyword stuffing.
- **first_person_ratio(text)**: Calculates the ratio of first-person pronouns (e.g., “I”, “me”, “we”) to total words. A lack of personal perspective may signal inauthentic content.
- **avg_sentence_length(text)**: Calculates the average number of words per sentence. Very short or overly long sentences may signal unnatural structure.
- **rating_sentiment_mismatch(text, rating)**: Compares the sentiment of the text with the numeric rating. If

sentiment is positive and rating ≤ 2 , or vice versa, it indicates a mismatch.

B. Feature Selection

Feature selection is a key step in enhancing the performance and interpretability of fake review detection models. Using hypothesis testing and correlation analysis, we identify the most relevant features from a large set of potential indicators, such as linguistic patterns and statistical properties. This approach helps eliminate redundant or irrelevant features, improving model accuracy, reducing computational cost, and minimizing overfitting, ultimately leading to better generalization on unseen data.

1) Hypothesis Testing

Hypothesis testing is used to examine whether certain features differ significantly between real and fake reviews. Identifying these differences helps in understanding the patterns commonly found in fake reviews and supports the selection of useful features for building classification models. Features that show strong statistical differences are likely to be valuable for improving model performance.

a) Welch's T-Test for Numerical Features: The dataset was divided into two groups based on the review label: real reviews (label = 0) and fake reviews (label = 1). A two-sample Welch's t-test was applied to each numerical feature, suitable when variances and sample sizes differ.

Null Hypothesis (H_0): The mean of the feature is the same for both real and fake reviews.

Alternative Hypothesis (H_1): The mean of the feature differs between real and fake reviews.

A significance level of 0.05 was used. Features with $p\text{-value} < 0.05$ were considered statistically significant. As shown in Fig.1, most features—including review_length, unique_word_count, entropy, and short_word_ratio—exhibited strong statistical significance, indicating meaningful differences between the two groups.

b) Chi-Square Test for Categorical Feature: The categorical feature category_relevance was evaluated using a chi-square test to determine whether it is associated with the review label. As shown in Fig.1, the chi-square statistic was 15.34 with a p-value of 0.0001, indicating a statistically significant association. The figure also summarizes all t-test and chi-square test results used for hypothesis testing.

Null Hypothesis (H_0): category relevance is independent of the review label. **Alternative Hypothesis (H_1):** category relevance is dependent on the review label.

2) Correlation Analysis

a) Feature Correlation:

To identify potential redundancies and relationships among numerical features, a feature correlation matrix was computed. This helps in understanding which features are linearly dependent on each other and can cause multicollinearity in machine learning models. As shown in Fig. 2, strong

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--- Hypothesis Testing ---
T-test for review_length: t=25.04, p=0.0000
T-test for word_count: t=17.92, p=0.0000
T-test for unique_word_count: t=41.31, p=0.0000
T-test for avg_word_length: t=44.34, p=0.0000
T-test for sentiment_score: t=-12.37, p=0.0000
T-test for punctuation_count: t=24.36, p=0.0000
T-test for entropy: t=62.32, p=0.0000
T-test for star_rating: t=-0.58, p=0.5603
T-test for uppercase_ratio: t=-3.41, p=0.0006
T-test for exclamation_count: t=17.66, p=0.0000
T-test for question_count: t=22.62, p=0.0000
T-test for digit_count: t=20.47, p=0.0000
T-test for word_length_variance: t=1.03, p=0.3030
T-test for lexical_diversity: t=48.14, p=0.0000
T-test for short_word_ratio: t=-65.90, p=0.0000
T-test for repetition_ratio: t=-46.34, p=0.0000
T-test for first_person_ratio: t=-27.49, p=0.0000
T-test for avg_sentence_length: t=51.18, p=0.0000
T-test for rating_sentiment_mismatch: t=-14.25, p=0.0000
Chi-square for category_relevance: chi2=15.34, p=0.0001

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Fig. 1. Hypothesis Testing results including t-test and chi-square statistics for selected features.

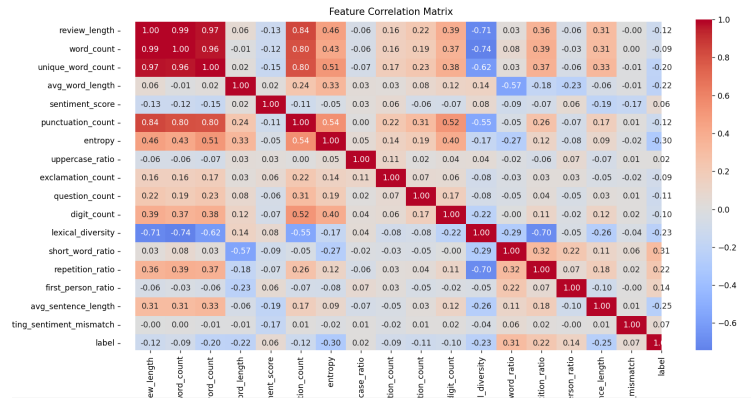


Fig. 2. Heat map showing pairwise correlation among numerical features.

correlations (closer to 1 or -1) can be observed between some features such as review_length, word_count, and unique_word_count. To complement this, we also calculated the correlation of each feature with the target variable (label), which is visualized in Fig.2. Table I lists the correlation values between each feature and the label and highlights the features that were dropped due to redundancy. This analysis helps in understanding which features are more predictive of whether a review is real or fake, with features like short_word_ratio, entropy, and repetition_ratio showing relatively higher correlation with the label. Based on the overall correlation matrix and feature-label correlation analysis, we identified and removed highly correlated features to reduce redundancy and prevent multicollinearity. Specifically, word_count and unique_word_count were dropped due to their high correlation with review_length, as also indicated at the bottom of Fig.2. These steps contribute to building a more robust and generalizable model by ensuring the final set of features is diverse and minimally interdependent.

TABLE I
FEATURE-LABEL CORRELATIONS AND FEATURES DROPPED

Feature	Correlation
review_length	-0.124
word_count	-0.089
unique_word_count	-0.201
avg_word_length	-0.215
sentiment_score	0.061
punctuation_count	-0.120
entropy	-0.296
uppercase_ratio	0.017
exclamation_count	-0.088
question_count	-0.112
digit_count	-0.101
lexical_diversity	-0.233
short_word_ratio	0.311
repetition_ratio	0.225
first_person_ratio	0.135
avg_sentence_length	-0.247
rating_sentiment_mismatch	0.071
Highly correlated features to drop: {‘word_count’, ‘unique_word_count’}	

C. Clustering

K-Means and Spectral Clustering algorithms were applied to the scaled training features within each product category to uncover latent groupings in the review data. Both algorithms are unsupervised learning techniques, meaning they operate without reference to the ground-truth labels (i.e., whether a review is real or fake) during the clustering process. Instead, they analyze the structure of the feature space to identify natural divisions based on similarity and distance metrics.

To further leverage the insights gained from these unsupervised methods, the resulting cluster assignments were incorporated as additional features into the supervised classification models. This approach allows the classifier to benefit from higher-level abstractions discovered by the clustering algorithms. By embedding these cluster labels as new categorical features, we aim to capture hidden patterns and complex relationships within the data that might not be apparent from the original features alone.

This strategy enriches the feature set and aids the model in understanding subtle behavioral or linguistic cues that may characterize fake or genuine reviews. In particular, clustering can group together reviews with similar stylistic or sentiment-based traits, potentially exposing deceptive patterns that traditional features may fail to isolate. Thus, the integration of K-Means and Spectral Clustering as feature extraction steps adds an additional layer of representation that enhances the overall performance and interpretability of the fake review detection system.

Spectral clustering is an unsupervised learning technique that identifies clusters in data by leveraging the eigenstructure of a similarity graph. Unlike traditional clustering methods that rely on geometric distance alone (e.g., k -means), spectral clustering captures the global structure of the data manifold, making it effective for detecting non-convex clusters.

In this work, the resulting cluster labels (e.g., 0 and 1) are not directly interpreted as fake or real reviews. Instead,

they serve as new features that encode structural similarities among data points. These features are fed into supervised models, helping them learn associations between graph-based groupings and the true class labels. This approach enhances model performance by capturing latent patterns that may not be obvious from raw features alone.

D. SMOTE: Synthetic Minority Oversampling Technique

SMOTE is a method used in machine learning to address imbalanced datasets, where one class has significantly more observations than the others. SMOTE generates synthetic samples of the minority class to create a more balanced dataset.

Within the category-specific modeling process, SMOTE is applied to the training data for each category to augment the dataset and enhance model performance. Although the overall dataset is balanced, SMOTE is utilized to generate synthetic review samples within each category, thereby expanding the feature space and exposing the models to a wider range of category-specific review characteristics. This process, implemented as the pipeline component `smote_pipeline`, uses SMOTE (SMOTE is `fit_resample` method), occurs after the data is grouped by category and split into training and testing sets, ensuring that only the training set is augmented. By augmenting the training data in this manner, the aim is to improve the generalization ability of the models within each category and assess their robustness to variations in review patterns.

E. Classification

1) Random Forest

This study uses the Random Forest model to classify reviews into real or fake based on the processed features. The training data ($X_{\text{train_final}}$) incorporates engineered linguistic and behavioral features as well as possible cluster label features. An ensemble of 100 decision trees is built (`n_estimators` = 100), each one: 1) fitted on a bootstrap sample of the data; and 2) trained on a random subset of features. With the aim to reduce overfitting, the trees’ maximum depth is restricted to 5 (`max_depth` = 5). The final prediction is determined by a majority vote of the 100 trees.

2) XGBoost (Extreme Gradient Boosting)

The current study leverages XGBoost as a supervised classification method for the detection of fake reviews. It uses engineered features from the review data to learn fine-grained patterns that might indicate real or fake reviews, such as the use of repeated language, unusual timing, or inconsistency with ratings. The model uses gradient boosting and includes regularization through parameters `reg_lambda` and `reg_alpha` to prevent overfitting. It also supports column subsampling, making it scalable and effective when handling large feature sets derived from textual and behavioral attributes.

3) *Logistic Regression*: Logistic Regression is employed in this study as a baseline model to classify reviews as real or fake. It assigns linear weights to the engineered features and uses the sigmoid function to estimate the probability of a review being fake. For instance, it may identify patterns such as excessively short reviews with a high exclamation mark count as indicative of fake reviews. Due to its simplicity and interpretability, Logistic Regression serves as a useful benchmark for comparison against more complex models. Model performance is assessed using the F1 score to account for potential class imbalance between real and fake reviews

F. Cross-Validation

Cross-validation is one of the main one-parameter techniques of machine learning that are used to measure the accuracy and generalizability of a model. We applied 5-fold cross-validation ($k = 5$) in this study to meticulously evaluate how well classifiers perform in distinguishing true or fake reviews categorized into a set of product categories.

Given k -fold cross-validation, the data set is first randomly split into k equal parts. In each iteration, one fold is used as the validation set, and all others, $k - 1$ of them, are aggregated for training. The process continues for k iterations, such that every data point has been exercised for validation exactly once. Averaging the performance metric—in our case, the F1-score—thus gives a reasonable estimate for an accurate model, while the standard deviation indicates stability, i.e., how much the results vary from fold to fold.

Some advantages of the 5-fold cross-validation are preventing overfitting due to validation on unseen data, being a better approach to use the whole data available for training (especially in cases where the data are limited or imbalanced), and ensuring fairness in how one compares the model with others. Cross-validation in our context is applied to each product category and clustering method (KMeans and Spectral), thus preserving generalizability and consistency throughout.

More importantly, cross-validation promotes good practices in hyperparameter tuning and model selection. By looking at performance across folds, we zero in on models that show consistent and reliable performance, whose results are not defined by some lucky split and that therefore could be robustly used in deployment scenarios in the real world for fake review detection.

V. RESULTS

A. Performance Comparison Across Product Categories

K-Means and Spectral Clustering were used on each product category to generate cluster-based features that were added to five classifiers: Logistic Regression, Random Forest, Gradient Boosting, and XGBoost. Cross-validation and test results are summarized below.

Table-2 presents the accuracy and F1-scores of different classification models (Random Forest, Logistic Regression, and XGBoost) evaluated using both KMeans and Spectral clustering across various product categories. The category numbers in the table correspond to specific product

TABLE II
BEST MODEL AND PERFORMANCE METRICS BY CATEGORY

Category No	Random Forest Accuracy (Kmeans)	Logistic Regression Accuracy (Kmeans)	XG Boost Accuracy (Kmeans)	Random Forest Accuracy (Spectral)	Logistic Regression Accuracy (Spectral)	XG Boost Accuracy (Spectral)	Cross Validation
Category1	0.852	0.904	0.904	0.863	0.906	0.890	Verified
Category2	0.866	0.925	0.884	0.866	0.922	0.884	Verified
Category3	0.883	0.920	0.902	0.883	0.924	0.902	Verified
Category4	0.872	0.919	0.895	0.872	0.903	0.895	Verified
Category5	0.871	0.930	0.913	0.866	0.931	0.913	Verified
Category6	0.879	0.908	0.897	0.879	0.903	0.897	Verified
Category7	0.875	0.921	0.895	0.873	0.878	0.887	Verified
Category8	0.880	0.928	0.919	0.871	0.851	0.919	Verified
Category9	0.860	0.904	0.873	0.867	0.886	0.863	Verified
Category10	0.848	0.916	0.893	0.848	0.797	0.893	Verified

domains as follows: Category 1 – *Books_5*, Category 2 – *Clothing_Shoes_And_Jewelry_5*, Category 3 – *Electronics_5*, Category 4 – *Home_and_Kitchen_5*, Category 5 – *Kindle_Store_5*, Category 6 – *Movies_and_TV_5*, Category 7 – *Pet_Supplies_5*, Category 8 – *Sports_and_Outdoors_5*, Category 9 – *Tools_and_Home_Improvement_5*, and Category 10 – *Toys_and_Games_5*.

As shown in Table V, Logistic Regression with either clustering approach—Spectral Clustering (LR(S)) or K-Means (LR(K))—consistently achieved the best and most stable results across all categories.

TABLE III
BEST PERFORMING MODEL AND ACCURACY PER CATEGORY

Cat	Best Model	Test Accuracy
Category 1	LR(S)	0.906
Category 2	LR(K)	0.926
Category 3	LR(S)	0.924
Category 4	LR(K)	0.919
Category 5	LR(S)	0.931
Category 6	LR(K)	0.909
Category 7	LR(K)	0.921
Category 8	LR(K)	0.928
Category 9	LR(K)	0.904
Category 10	LR(K)	0.916

B. Model Evaluation with User-Provided Input

To test practical performance, reviews with star ratings were manually input to verify the model's ability to detect fake reviews. The results show moderate performance, with some categories showing better results for fake review detection than real ones, as summarized in Table IV (Table 4):

Table V shows that Logistic Regression consistently achieved high precision (0.89–0.93) across all categories. LR(S) performed best in categories 1, 3, and 5, while LR(K)

TABLE IV
CLASSIFICATION RESULTS OF REAL AND FAKE REVIEWS ACROSS
DIFFERENT CATEGORIES

Category No	Real detected as real	Fake detected as fake
Category 1	17/25 (68%)	15/25 (60%)
Category 2	17/25 (68%)	19/25 (76%)
Category 3	18/25 (72%)	15/25 (60%)
Category 4	13/20 (65%)	14/20 (70%)
Category 5	12/20 (60%)	14/20 (70%)
Category 6	16/20 (80%)	15/20 (75%)
Category 7	15/20 (75%)	10/20 (50%)
Category 8	14/20 (60%)	11/20 (55%)
Category 9	11/20 (55%)	12/20 (60%)
Category 10	14/20 (60%)	15/20 (75%)

TABLE V
PRECISION

Cat	Best Model	Precision
Category 1	LR(S)	0.91
Category 2	LR(K)	0.90
Category 3	LR(S)	0.91
Category 4	LR(K)	0.92
Category 5	LR(S)	0.93
Category 6	LR(K)	0.91
Category 7	LR(K)	0.92
Category 8	LR(K)	0.93
Category 9	LR(K)	0.90
Category 10	LR(K)	0.89

dominated the remaining categories, highlighting its broader adaptability.

Overall, the models showed solid ability to detect fake reviews but moderate performance on real reviews, suggesting areas for further improvement.

VI. CONCLUSION

The paper discusses a robust mechanism for detecting spurious (fake) reviews by incorporating experience in feature engineering, statistical hypothesis testing, and machine learning. A variety of features from the review texts, spanning from linguistic ones, such as text length, lexical diversity, and sentiment, to structural ones, including punctuation usage and relevance to product category are extracted. TF-IDF-based features helped to further differentiate the genuine ones from the deceptive reviews based on unique patterns of word usage. The univariate statistical hypothesis testing procedure, relying on t-test and chi-square test, was useful for selecting those features that significantly differ between fake and genuine ones. With this approach, the models are expected to base predictions on meaningful variables thus ensuring better interpretability and prediction accuracy. The results consistently revealed that fake reviews tend to be shorter, less diverse in vocabulary, and usually unrelated to the product category, thereby adding merit to the context-aware features. By testing different ML models with k-fold cross-validation, the robustness/transience of the approach across various product categories is ensured. The proposed method is observed to work well to identify fake reviews yet can be reapplied (with slight modifications) in other domains. This work also showed how combining statistical rigor with ML leads to systems that are explainable

and scalable for the detection of fake reviews in a broader theme as in trust and content authenticity analysis.

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